

Arun Keshav Vaithianathan

(409) 202-9289 | akvaithi.tech@gmail.com | [linkedin.com/in/akvaithi](https://www.linkedin.com/in/akvaithi) | github.com/akvaithi

EDUCATION

Texas A&M University

August 2024 – May 2028

Bachelor of Science in Chemical Engineering; Minor in Mathematics

College Station, TX

- Cumulative GPA: 3.918
- **Honors & Programs:** Craig and Galen Brown Engineering Honors, Aggies in Tech Business Certificate Program
- **Awards:** Dean's List, Chemical Engineering Department Scholarship, Lechner Scholarship, Aggies in Tech Scholarship

EXPERIENCE

Undergraduate Researcher

February 2026 – Present

Texas A&M University College of Engineering

College Station, TX

- Investigating catalytic conversion of petroleum coke into battery-grade graphite for sustainable, low-cost anode materials
- Executing material synthesis via high-energy ball milling and iron catalyst integration, operating high-temperature tube furnaces to drive graphitization
- Characterizing synthesized samples utilizing X-ray Diffraction (XRD) to evaluate structure and validate graphitization degree

Control Systems Engineering Lead

May 2025 – Present

Texas A&M University, Oceanus ThinkTank, MATE ROV

College Station, TX

- Engineered a closed-loop feedback control system (PID) utilizing Python and ArduSub firmware to maintain vehicle stability under dynamic fluid conditions
- Designed a Distributed Control System (DCS) architecture by integrating a Raspberry Pi 5 and Blue Robotics Navigator flight controller to manage 6-DOF IMU data and thruster output
- Conducted a techno-economic trade study on single-board computers (SBCs), selecting a high-bandwidth architecture to support photogrammetry and turbidity sensing requirements
- Developed a real-time computer vision tool that ingests a live underwater RTSP feed over Ethernet, using variance-based frame-acquisition logic and Apple's RealityKit to generate high-fidelity 3D reconstructions for spatial analysis

Project Co-founder

January 2026 – Present

Aggie Research Finder

College Station, TX

- Architected and deployed a centralized discovery platform indexing over 1,770 Texas A&M faculty members, streamlining undergraduate research matching and scaling to 3,000+ active users
- Engineered a high-performance web architecture utilizing modern frontend frameworks and optimized databases to handle rapid, multi-attribute querying across massive institutional datasets
- Designed intuitive data filtering algorithms to parse faculty research interests, helping peers easily identify relevant labs based on specific engineering and scientific disciplines

Lead Developer & Creator

May 2026

Tube Furnace Program Calculator

College Station, TX

- Architected and deployed a single-page web application to automate the calculation of complex, multi-stage thermal profiles up to 1,600°C for advanced materials synthesis and laboratory workflows
- Engineered a logic-driven mathematical interface that dynamically generates precise setpoint tables, factoring in user-defined ramp rates, preheat hold durations, and cooling markers
- Eliminated manual configuration errors for laboratory personnel, ensuring reproducible furnace programming and high-fidelity thermal cycles for critical research experiments; deployed via Vercel

Texas Exchange for Energy & Climate Entrepreneurship (TEX-E) Fellow

May 2025 – May 2026

Greentown Labs | Ecosystem Builders Committee Co-lead

Houston, TX

- Co-lead the Ecosystem Builders Committee, collaborating with partner universities to promote the TEX-E fellowship, organize renewable energy events, and connect students with industry leaders
- Assessed early-stage technologies in carbon capture, hydrogen production, and low-carbon fuels for process feasibility and scalability within industrial applications

- Launched and grew the TEX-E Instagram to over 400 followers by translating complex Chem-E insights into engaging bite-sized visuals, advancing leadership in energy-transition storytelling
- Leveraged fellowship events to network with founders, VCs, and plant engineers, funneling their insights into content that fuels new partnerships between A&M researchers and Greentown Labs portfolio companies

Aggies Invent: ExxonMobil Engineered for Safety Team Lead

November 2025

Texas A&M University, Meloy Program

College Station, TX

- Awarded 3rd Place Overall for designing a SIL-rated safety interlock system utilizing Two-out-of-Two logic and ISO 13849-1 standards to prevent struck-by incidents in high-risk zones
- Engineered a hardware-agnostic E-stop integration utilizing SAE J1939 protocols, triggering automatic shutdowns via Edge AI and UWB proximity sensors upon detecting dynamic buffer zone breaches
- Designed a scalable deployment strategy distinguishing between Upstream (mobile magnet-mount) and Downstream (fixed network) infrastructures, specifically addressing HazLoc (Hazardous Location) constraints

SEC Ignite Engineering Design Competition

October 2025

Texas A&M University

College Station, TX

- Awarded 2nd Place Overall for developing a comprehensive safety solution for Chart Industries' design challenge
- Engineered enhanced fail-safes and risk mitigation strategies for Air Cooled Heat Exchanger (ACHE) systems

Information Technology Support Intern

August 2024 – May 2025

Texas A&M University Athletics (Kyle Field)

College Station, TX

- Managed and optimized a high-density network of 350+ Point-of-Sale (POS) systems, including 200+ Oracle MICROS terminals and 150+ enterprise iPads, across a 102,000+ capacity stadium
- Ensured 100% operational uptime during high-traffic SEC game days by proactively monitoring hardware infrastructure and diagnosing real-time network connectivity issues
- Collaborated with cross-functional technical teams to rapidly resolve critical system faults under high-pressure, time-sensitive live event conditions

PVFA Student Videographer

August 2025 – January 2026

TAMU School of Performance, Visualization & Fine Arts

College Station, TX

- Produced, captured, and edited high-quality video assets to document and promote university-sanctioned performances, exhibitions, and academic initiatives
- Operated professional cinema cameras, audio capture equipment, and lighting rigs in diverse environmental conditions ranging from live stage productions to studio interviews
- Managed post-production workflows, executing color grading, multi-cam editing, and asset delivery aligned with institutional branding guidelines

Machine Learning Predictive Modeling Developer

September 2024

Texas A&M University, TAMUHack

College Station, TX

- Developed a TensorFlow-based neural network backend in Python to accurately predict horse race placements based on historic performance metrics and data preprocessing
- Collaborated to integrate the trained model with a JavaScript web application for real-time inference and user interaction

Summer Robotics Camp Instructor

May 2025 – August 2025

Robotics Education

Houston, TX

- Instructed and mentored students in foundational robotics, programming, and engineering design principles through hands-on projects and team-based competitions

Lifeguard

June 2023 – October 2023

City of Pearland Natatorium

Pearland, TX

- Supervised swimming pool areas and communicated with various team members and patrons to ensure patrons' safety
- Earned American Red Cross CPR Certification

LEADERSHIP & EXTRACURRICULAR

Program Cohort Member

August 2025 – Present

Aggies in Tech — TAMU Mays Business School

College Station, TX

- Selected for a highly competitive, cross-disciplinary cohort bridging engineering and business disciplines through specialized coursework in technology commercialization, venture capital, and corporate strategy
- Participated in an intensive Silicon Valley executive residency, engaging with leadership and engineering teams at Google, Meta, LinkedIn, Visa, and Perplexity to analyze emerging market trends
- Collaborated with Silicon Valley startup founders and GNC specialists to evaluate technical overlaps between commercial drone flight software (5G architectures) and underwater robotics control systems

Pan-Honors Freshman Leadership Organization (PHFLO) Outreach Team

May 2025 – Present

Texas A&M University

College Station, TX

- Managed recruitment, visibility, and public relations, overseeing social media, digital announcements, flyers, and physical marketing to grow the organization's public presence
- Designed the official Pan-Honors Instagram logo and maintained consistent branding to ensure target audiences were engaged with upcoming programs and application cycles

AIChE Publicity & Outreach Committee Member

August 2024 – Present

Texas A&M University, AIChE

College Station, TX

- Operated the official chapter Instagram account, increasing student engagement by promoting professional development opportunities, departmental information, and outreach events
- Designed custom merchandise, including chapter t-shirts and sweatshirts, to strengthen visibility and recruit new chemical engineering student members

Digital Media Creator, Photographer, & Videographer

December 2020 – Present

Arun Vaithianathan Photography

Houston, TX

- Scaled a technical education YouTube channel to 2.2k+ subscribers and 1M+ views by producing, scripting, and editing high-yield video tutorials on Python automation, Linux deployment, and self-hosted server architectures
- Directed end-to-end commercial photography and videography workflows for 80+ client engagements, managing logistics, client relations, and advanced post-processing pipelines
- Monetized creative assets through strategic sponsorships and the YouTube Partner Program, establishing a self-sustaining business model that completely offsets 100% of operational and production overhead

FIRST Robotics Competition Team 5414 Programmer & Media Lead

August 2021 – May 2024

Pearadox FRC Facility

Pearland, TX

- Programmed autonomous routines and machine vision auto-aiming in Java (WPILib), securing 6 blue banners and advancing to the FRC World Championship for three consecutive years
- Managed media team and YouTube channel, acquiring over \$15,000 in funding and grants across 3 years

SKILLS

Process Simulation & Engineering: Aspen HYSYS, LOPA, HAZOP, SIL Assessment, PID Loop Tuning, DCS Architecture, SAE J1939 Protocols, LabView, Autodesk Fusion 360

Programming Languages: Python, Java, C++, HTML/CSS, JavaScript, Visual Basic for Applications (VBA)

Libraries & Frameworks: TensorFlow, NumPy, Pandas, Matplotlib, React, OpenCV, WPILib

Software & Tools: Linux/Unix, Git, Docker, Google Cloud Platform (GCP), VS Code, IntelliJ, Google Firebase, Adobe Creative Cloud, Microsoft Office Suite